Hot Plate Design

- A 150 mm × 55 mm 6061 aluminium plate is heated by nine vertical 230 V cartridge heaters in two zones (3 in the centre, 6 in an outer ring), all at least 15 mm below the surface so the thick metal smooths out hot spots.
- Each zone has its own 4-wire PT100 read by a MAX31865, and a Pico runs one PID loop per zone through zero-cross SSRs, which sets the centre-to-edge power ratio that cancels edge heat loss.
- The plate sits on low-contact standoffs with ceramic insulation, a metal draft shield, and a high-temperature limiter on the mains line.
- Accuracy comes from calibration, and repeatability from a fixed process (same ramp, soak rule, start temperature and wafer position on 3 pins).
- Flatness comes from heat-cycling before a final fly-cut, and everything is checked with an instrumented wafer, 10 back-to-back runs and a dial indicator.